Training Recommender Systems with Biasing Data: A Comparison of SVD and NMF

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1. Three Datasets

Recommenders learn from user-item ratings to rank unseen items. We study how added synthetic ratings change category exposure for real users.

Books: Goodbooks-10K

53,424 real users

9,415 items

5,815,569 observed ratings

13 assigned genres; books may have two labels.

Movies: MovieLens 100K

943 real users

1,682 items

100,000 observed ratings

Release decades are the target categories.

Mobile Apps: MobileRec

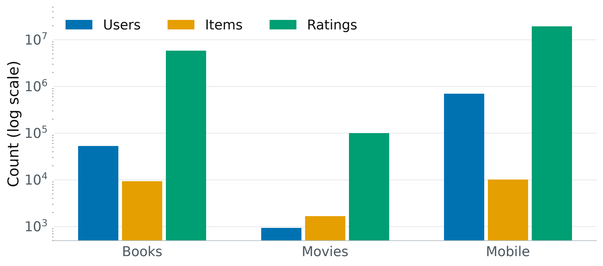
700,111 real users

10,173 items

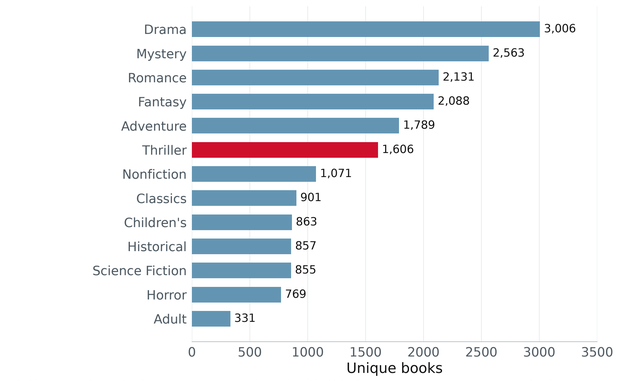
19,297,019 observed ratings

48 original categories; Shopping is the example target.

Dataset Scale



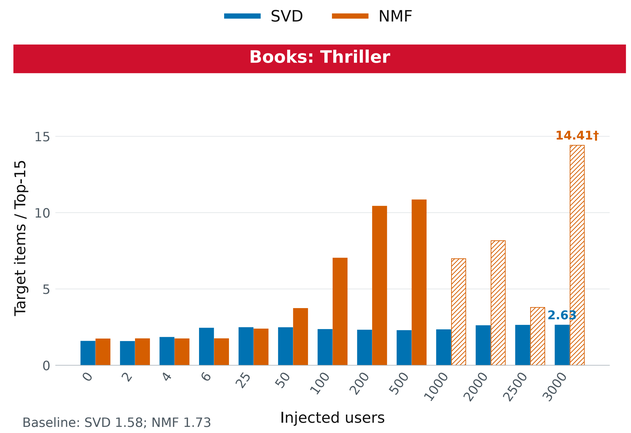
Book Genres After Labeling



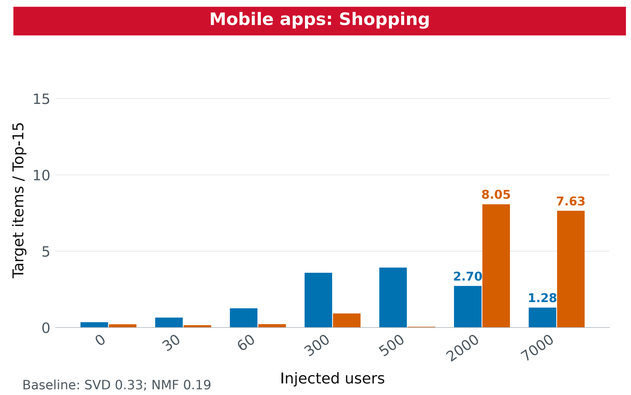
Ratings are 1-5. Book genres were assigned with GPT-3.5; exposure measurements depend on these labels.

2. Results Across All Three Datasets

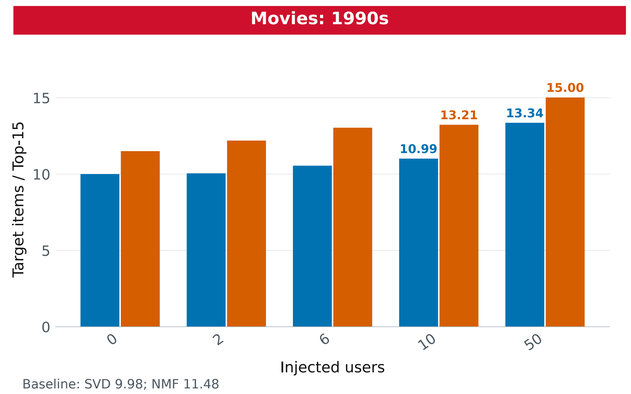
Mean number of target-category items in each real user's Top-15. Zero is the baseline; each point is a separate training run.



Baseline: SVD 1.58; NMF 1.73



Baseline: SVD 0.33; NMF 0.19



Baseline: SVD 9.98; NMF 11.48

Full displayed ranges

Books: 0-3,000; mobile: 0-7,000; movies: 0-50. Injection counts are categorical. Rating budgets differ across datasets.

† Hatched book NMF bars are raw outputs from checkpoints that failed the factor non-negativity check (1,000, 2,000, 2,500 and 3,000). The n=500 comparison passes this check.

3. Understanding SVD and NMF

Both models predict ratings with:

$$\hat{r}_{ui} = \mu + b_u + b_i + p_u^\top q_i$$

μ	Global mean of the training ratings.
b_u	Learned user bias: one offset per user.
b_i	Learned book bias: one offset per book.
$p_u^\top q_i$	Dot product of the learned user and book vectors.

SVD: Singular Value Decomposition

User and item vectors can contain positive and negative values. Additive gradient updates adjust the factors and biases to reduce rating prediction errors.

NMF: Non-Negative Matrix Factorization

Factors are intended to remain non-negative and are updated multiplicatively. User and item biases are signed and updated additively.

The Same Injection Rule

Each fake user rates all 1,606 Thriller books 5 and all 7,809 other books 1. No books are added.

At n=500: 4,707,500 new ratings, equal to 80.9% of the original book rating count.

Book Case Study: n = 500

Training on all users: 53,424 real + 500 fake. Below, each row is a mean over observed rating pairs.

Common global mean: $\mu = 2.918$

Model / books	b_u	b_i	$p_u^\top q_i$	$\hat{r}$	r
SVD / T	0.210	0.457	0.839	**4.425**	4.436
NMF / T	-0.890	2.216	0.062	**4.305**	4.436
SVD / other	0.211	0.150	-0.648	**2.631**	2.640
NMF / other	-0.861	-1.404	1.721	**2.374**	2.640

T = Thriller; other = non-Thriller.

Different Components, Similar Score Trends

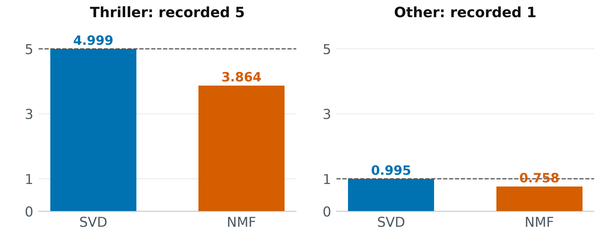
SVD: book bias and dot products both favor Thriller over other books. NMF: book bias creates a larger separation, while dot products partially offset it.

The mean user bias is positive in SVD and negative in NMF. Adding all four components gives the final predicted rating.

Shared settings: 8 factors; 180 epochs; seed 42.

4. Why Do the Outputs Differ? (Books, n=500)

First, inspect only the fake users' ratings from the same models trained on all real and fake data. Their desired ratings are exactly 5 and 1.



Training MAE (Thriller / other):
SVD: 0.012 / 0.006 NMF: 1.136 / 0.244

SVD uses positive and negative dot products. NMF's dot products are almost zero for these fake profiles; biases supply most of their score separation.

Then Predict for Real Users

No parameters are updated. Rank unrated books for each real user. The global mean and that user's bias are the same for every candidate, so ranking depends on:

$$b_i + p_u^\top q_i$$

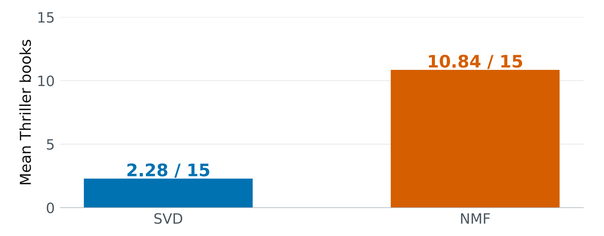
Thriller minus non-Thriller mean score:

NMF: $3.096 - 2.635 = +0.461$

SVD: $0.296 - 0.396 = -0.100$

The first term is the book-bias advantage; the second is the opposing dot-product difference. The NMF bias advantage remains after this compensation.

Observed Top-15 Recommendations



Conclusion

SVD fits the training ratings more closely. NMF gives Thriller stronger real-user exposure at n=500, alongside a larger book-bias separation that vectors only partly offset.

Exposure is not held-out accuracy. These results describe the tested models and injection policy.

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